INCH-POUND
MIL-DTL-22200/1J(SH)
14 February 2025
SUPERSEDING
MIL-DTL-22200/1H(SH)
w/AMENDMENT 2
02 July 2019

DETAIL SPECIFICATION SHEET

ELECTRODES, WELDING, MINERAL COVERED, IRON-POWDER, LOW-HYDROGEN MEDIUM AND HIGH TENSILE STEEL, AS-WELDED OR STRESS-RELIEVED WELD APPLICATION

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-22200.

1. SCOPE

1.1 Scope. This specification covers iron-powder, low-hydrogen types of mineral-covered electrodes, for the fabrication and repair welding of low and medium carbon and high tensile steels for as-welded and stress-relieved applications.

1.2 Classification. Iron-powder, low-hydrogen electrodes are furnished in the types, classes, and sizes as specified in table I (see 6.2).

TABLE I. Types, classes, positions of welding, and sizes.

Type 1/, 2/	Class	Positions of welding	Sizes (diameter, inch)
MIL-7018-M MIL-8018-C3 MIL-8018-C3SR	1	All	3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 7/32, 1/4
	3	Flat	5/16
MIL-9018-M MIL-11018-M	1	All	3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 7/32, 1/4

NOTES:

1/ Type classifications of electrodes will be determined according to the capability of the electrode to meet the tests as specified herein, when used in the positions of welding as specified in this table and section 4 herein. An electrode will be approved under only one type of classification, and an electrode will not be approved if it differs in type from that stated in the manufacturer's request for test. The position description indicates the maximum size qualified under the specification. However, reference should be made to the Qualified Products List (QPL) for the selection of electrodes for all positions of welding.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 3439

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

TABLE I. Types, classes, positions of welding, and sizes – Continued.

NOTES (continued):

2/ The suffix RC added to the MIL-type pertains to restriction of the copper content (see table IV, note 4). All requirements, tests, and acceptance criteria remain otherwise unchanged.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-22200	-	Electrodes, Welding, Covered: General Specification for
MIL-S-22698	-	Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVSEA TECHNICAL PUBLICATIONS

T9074-BD-GIB-010/0300	-	Base Materials for Critical Applications: Requirements for Low Alloy Steel Plate, Forgings, Castings, Shapes, Bars, and Heads of HY-80/100/130 and HSLA-80/100

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN WELDING SOCIETY (AWS)

AWS A5.5/A5.5M	-	Specification for Low-Alloy Steel Electrodes for Shielded Metal Arc Welding

(Copies of this document are available online at www.aws.org.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

ASTM INTERNATIONAL

ASTM A36/A36M - Standard Specification for Carbon Structural Steel

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specification exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Electrodes furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Electrodes. Electrodes furnished under this specification shall be in accordance with the requirements of MIL-DTL-22200 and as specified herein.

3.3 Covering composition. Except as specified in 3.3.1, 3.3.2, and 3.3.3 and subject to the requirements of MIL-DTL-22200, the chemical composition of the electrode covering material is at the manufacturer's discretion.

3.3.1 Covering moisture. The moisture content of the electrode coverings shall not exceed the percent by weight values as specified in table II.

TABLE II. Covering moisture.

Type	Maximum covering moisture (wt. %)	
	Condition A 1/	Condition B 2/, 3/
MIL-7018-M	0.15	0.25
MIL-8018-C3	0.15	0.25
MIL-8018-C3SR	0.15	0.25
MIL-9018-M	0.15	0.25
MIL-11018-M	0.15	0.25

NOTES:
1/ Condition A - When removed from the manufacturer's sealed container.
2/ Condition B - After 9 hours (minus 0, plus 30 minutes) of exposure at 80 percent relative humidity and 80 °F as specified in 4.5.
3/ For receipt inspection by the contracting activity, the maximum water content shall be not greater than 0.29 percent.

3.3.2 Diffusible hydrogen level. For MIL-7018-M and MIL-11018-M electrodes, the diffusible hydrogen levels in milliliters per one hundred grams (mL/100g) of deposited weld metal shall not exceed the following values for welds deposited with electrodes removed from sealed manufacturer's containers:

a. For electrodes ⅛-inch size and smaller, the maximum average value is 3.2 mL/100g and the maximum single value is 4.0 mL/100g.

b. For electrodes 5⁄32-inch size and larger, maximum average value is 3.5 mL/100g and the maximum single value is 4.3 mL/100g.

3.3.3 Total iron content. The total iron content of the covering, including any combined iron in addition to metallic iron powder, shall be not less than 15 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

3.4 Groove welds.

3.4.1 Mechanical properties of deposited weld metals. The mechanical properties of weld metal deposited in groove joints shall be as specified in table III. When specified for conformance inspection (see table VIII and 6.2) and as required for qualification inspection (see table VII), stress-relief shall be in accordance with the following:

a. MIL-7018-M, MIL-8018-C3, and MIL-8018-C3SR. For MIL-7018-M, the temperature shall be 1,150±25 °F, held at this temperature for 1 hour, and then reduced to below 500 °F by furnace cooling at a maximum rate of 200 °F per hour. For MIL-8018-C3 and MIL-8018-C3SR, the temperature shall be 1,150±25 °F, held at this temperature for 2 hours, and then reduced to below 500 °F by furnace cooling at a maximum rate of 200 °F per hour.

b. MIL-8018-C3SR. The temperature shall be 1,150±25 °F, held at this temperature for a minimum of 50 hours, and then reduced to below 600 °F by furnace cooling at a maximum rate of 10 °F per hour.

c. MIL-9018-M and MIL-11018-M. The temperature shall be 1,025±25 °F, held at this temperature for 1 hour, and then reduced to below 500 °F by furnace cooling at a maximum rate of 200 °F per hour.

TABLE III. Mechanical properties of deposited weld metal. 1/

Tensile						
Electrode type	Ultimate tensile strength		Yield strength 2/		Elongation 3/	
	As-welded (psi) (minimum)	Stress-relieved (psi) (minimum)	As-welded 4/ (psi) range	Stress-relieved (psi) (minimum)	As-welded (percent) (minimum)	Stress-relieved (percent) (minimum)
MIL-7018-M	5/	---	53,000 to 75,000	51,000	24	24
MIL-8018-C3	80,000	---	68,000 to 80,000	65,000	24	24
MIL-8018-C3SR	---	80,000	---	65,000	---	24
MIL-9018-M	90,000	---	78,000 to 90,000	75,000	24	24
MIL-11018-M	5/	---	98,000 to 110,000	95,000	20	20

Charpy V-notch impact				
Type	As-welded		Stress-relieved	
	Energy min. avg. (ft-lbs)	Temp. (°F)	Energy min. avg. (ft-lbs)	Temp. (°F)
MIL-7018-M 6/	50	minus 20	35	minus 20
MIL-8018-C3 7/	20	minus 20	20	minus 20
MIL-8018-C3SR 7/	---	---	20	minus 20
MIL-9018-M 7/	20	minus 60	20	minus 60
MIL-11018-M 7/	35	minus 60	20	minus 60

NOTES:

1/ These are the minimum and maximum averages determined by testing all specimens required from each qualification or conformance test weld.

2/ Yield strength shall be determined by the offset method (offset = 0.2 percent).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

TABLE III. Mechanical properties of deposited weld metal – Continued. 1/

NOTES (continued):

3/ Percent reduction in area shall be reported for information only.

4/ The upper value of the yield strength may be 5,000 pounds per square inch (psi) higher than the indicated value for the 3/32-inch size electrode.

5/ The ultimate tensile strength shall be recorded for information only.

6/ At the test temperature as specified in this table, ±3 °F, the average value of five tests shall be equal to or greater than the minimum average value specified. No more than two specimens may have values below the minimum average specified, but each specimen must have a value greater than or equal to 20 foot-pounds (ft-lbs).

7/ Impact properties for the five specimens for each test condition shall be obtained at test temperatures as specified in this table, ±3 °F. The lowest impact value thus obtained together with the highest value shall be disregarded for this test. Two of the three remaining values shall be equal to or greater than that specified herein. One of the three remaining values may be up to 5 ft-lbs less than that specified herein. The computed average value of the three remaining values shall be equal to or greater than that specified herein.

3.4.2 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table IV.

TABLE IV. Chemical composition of deposited weld metal. 1/

Element	Chemical composition (percentage)			
	MIL-7018-M	MIL-8018-C3 MIL-8018-C3SR	MIL-9018-M	MIL-11018-M
Carbon	0.10	0.12	0.10	0.07 2/
Manganese	0.40 to 1.60	0.40 to 1.25	0.60 to 1.25	1.30 to 1.80
Silicon	0.80	0.80	0.80	0.60
Phosphorus	0.030	0.030	0.030	0.030
Sulfur	0.020	0.030	0.030	0.030
Chromium	0.15	0.15	0.15	0.40 3/
Nickel	0.25	0.80 to 1.10	1.40 to 1.80	1.25 to 2.50
Molybdenum	0.35	0.35	0.35	0.25 to 0.50
Vanadium	0.05	0.05	0.05	0.05
Copper	4/	4/	4/	4/
Boron	5/	---	---	5/

NOTES:

1/ Single values are maximum percentages.

2/ On sizes of MIL-11018-M electrodes over 3/16-inch diameter, the maximum carbon content shall be 0.10 percent.

3/ When specified by the purchaser, a maximum of 0.20 percent chromium content may be required for control of hexavalent chromium in fumes (see 6.2).

4/ When specified (see 6.2), use of the suffix RC (e.g., MIL-11018-MRC) indicates that the maximum copper content of the deposited weld metal shall be 0.15 percent. The remaining basic compositional requirements remain unchanged for the specific MIL-type. Report copper content for information only for electrodes without the RC suffix.

5/ Report for information.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

3.5 Electrode lengths. The specified (see 6.2) nominal lengths of electrodes shall be in accordance with table V except for type MIL-8018-C3SR. Type MIL-8018-C3SR may be purchased in longer lengths when specified and when qualified in the longer lengths (see 6.2).

TABLE V. Electrode lengths.

Sizes (diameter, inch)	Electrode lengths (inches)
3/32	9, 12, or 14 1/
1/8, 5/32, 3/16	14
7/32, 1/4, 5/16	18

NOTE:
1/ As specified (see 6.2)

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2 and table VI).

b. Conformance inspection (see 4.3 and table VI).

4.2 Qualification inspection. Qualification inspection shall include the examinations and tests of MIL-DTL-22200 and the additional tests in table VI herein. MIL-8018-C3SR shall be tested with the longest electrode intended to be used (see 3.5).

TABLE VI. Required additional tests for qualification and conformance inspections. 1/

Examinations and tests	Procedures	Requirements	Qualification	Conformance
Covering moisture	MIL-DTL-22200 and 4.5 herein	3.3.1 herein	X (Table II, conditions A and B)	X (Table II, conditions A and B)
Total iron content in covering	MIL-DTL-22200	3.3.3 herein	X	---
Diffusible hydrogen 2/	MIL-DTL-22200	3.3.2 herein	X	X
Groove weld tests	See Tables VII and VIII		Table VII	Table VIII

NOTES:
1/ See 4.2 and 4.3 for additional requirements.
2/ Required for MIL-7018-M and MIL-11018-M types only.

4.3 Conformance inspection. Conformance inspection shall include the tests and examinations specified in MIL-DTL-22200 and table VI herein.

4.3.1 Lot. A lot of electrodes shall be in accordance with MIL-DTL-22200 for AWS classification C3.

4.3.2 Sampling for conformance inspection. Sampling for conformance inspection shall be in accordance with MIL-DTL-22200 for those tests and examinations required in MIL-DTL-22200 and for the tests and examinations specified in table VI.

4.3.3 Certification of conformance. A certificate of conformance shall be in accordance with MIL-DTL-22200.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

TABLE VII. Groove weld tests required for qualification inspection.

Weld test assembly	Electrode type	Electrode diameter							Polarity	Welding position	Tests						Test procedures	Requirements (as specified herein)
											NDT [1/]			Tensile [2/]	Charpy	Chemical analysis [3/]		
		3/32	1/8	5/32	3/16	7/32	1/4	5/16			VT	MT	RT					
Figure 1	MIL-7018-M	X	X	X	--	--	--	--	DCEP	Vertical	X	--	X	X	X	--	MIL-DTL-22200	3.4.1 and Table III As-welded and stress-relieved condition (3.4.1.a)
														--	--	X		3.4.2 and Table IV
Figure 1	MIL-7018-M	--	--	--	X	X	X	X	DCEP	Flat	X	--	X	X	X	--	MIL-DTL-22200	3.4.1 and Table III As-welded and stress-relieved condition (3.4.1.a)
														--	--	X		3.4.2 and Table IV
Figure 1	MIL-8018-C3	X	X	X	X	X	X	X	DCEP	Flat	X	--	X	X	X	--	MIL-DTL-22200	3.4.1 and Table III As-welded and stress-relieved condition (3.4.1.a)
														--	--	X		3.4.2 and Table IV
Figure 1	MIL-8018-C3	X	X	X	X	X	X	X	AC	Flat	X	--	X	X	X	--	MIL-DTL-22200	3.4.1 and Table III As-welded and stress-relieved condition (3.4.1.a)
														--	--	X		3.4.2 and Table IV
Figure 1	MIL-8018-C3SR	X	X	X	X	X	X	X	DCEP	Flat	X	--	X	X	X	--	MIL-DTL-22200	3.4.1 and Table III Stress-relieved conditions (3.4.1.a and 3.4.1.b)
														--	--	X		3.4.2 and Table IV
Figure 1	MIL-9018-M	X	X	X	X	X	X	--	DCEP	Flat	X	--	X	X	X	--	MIL-DTL-22200	3.4.1 and Table III As-welded and stress-relieved condition (3.4.1.c)
														--	--	X		3.4.2 and Table IV
Figure 1	MIL-11018-M	X	X	X	X	X	X	--	DCEP	Flat	X	X	X	X	X	--	MIL-DTL-22200	3.4.1 and Table III As-welded and stress-relieved condition (3.4.1.c)
														--	--	X		3.4.2 and Table IV

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

TABLE VII. Groove weld tests required for qualification inspection – Continued.

Weld test assembly	Electrode type	Electrode diameter							Polarity	Welding position	Tests						Test procedures	Requirements (as specified herein)
											NDT 1/			Tensile 2/	Charpy	Chemical analysis 3/		
		3/32	1/8	5/32	3/16	7/32	1/4	5/16			VT	MT	RT					
Figure 2	MIL-8018-C3	X	X	X	--	--	--	--	DCEP	Vertical	X	--	X	--	--	--	MIL-DTL-22200	As-welded and stress-relieved condition (3.4.1.a)
Figure 2	MIL-8018-C3SR	X	X	X	--	--	--	--	DCEP	Vertical	X	--	X	--	--	--	MIL-DTL-22200	Stress-relieved condition (3.4.1.a and 3.4.1.b)
Figure 2	MIL-9018-M MIL-11018-M	X	X	X	--	--	--	--	DCEP	Vertical	X	--	X	--	--	--	MIL-DTL-22200	As-welded and stress-relieved condition (3.4.1.c)

NOTES:

1/ All NDT inspection (Visual Testing [VT], Radiographic Testing [RT], and Magnetic Particle [MT]) of weld test assemblies shall be performed to the procedures and requirements of MIL-DTL-22200. Acceptance criteria shall be in accordance with MIL-DTL-22200.

2/ Two tensile tests shall be prepared and tested, and includes ultimate, yield, elongation, and reduction in area (see table III).

3/ If chemical analysis is not determined using chemical analysis test pads welded as specified in MIL-DTL-22200, chemical analysis shall be determined from milling sample taken from the centerline of the broken tensile specimens or drillings, or by optical emission spectroscopy on any sample that is taken and analyzed at the depth of the tensile specimen centerline. In event of variances with composition requirements or conflicting results by testing activities, analysis shall be verified by procedures as specified in MIL-DTL-22200.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

TABLE VIII. Groove weld tests required for conformance inspection.

Weld test assembly	Electrode type	Electrode diameter							Polarity	Welding position	Tests						Test procedures	Requirements (as specified herein)
											NDT [1/]			Tensile [2/]	Charpy	Chemical analysis [3/]		
		3/32	1/8	5/32	3/16	7/32	1/4	5/16			VT	MT	RT					
Figure 1	MIL-7018-M	X	X	X	--	--	--	--	DCEP	Vertical	X	--	X	X	X	--	MIL-DTL-22200	As-welded condition (3.4.1 and Table III) Stress-relieved condition [4/]
														--	--	X		3.4.2 and Table IV
Figure 1	MIL-7018-M	--	--	--	X	X	X	X	DCEP	Flat	X	--	X	X	X	--	MIL-DTL-22200	As-welded condition (3.4.1 and Table III) Stress-relieved condition [4/]
														--	--	X		3.4.2 and Table IV
Figure 1	MIL-8018-C3	X	X	X	X	X	X	X	DCEP	Flat	X	--	X	X	X	--	MIL-DTL-22200	As-welded condition (3.4.1 and Table III)
														--	--	X		3.4.2 and Table IV
Figure 1	MIL-8018-C3SR	X	X	X	X	X	X	X	DCEP	Flat	X	--	X	X	X	--	MIL-DTL-22200	Stress-relieved condition [5/]
														--	--	X		3.4.2 and Table IV
Figure 1	MIL-9018-M	X	X	X	X	X	X	--	DCEP	Flat	X	--	X	X	X	--	MIL-DTL-22200	As-welded condition (3.4.1 and Table III) Stress-relieved condition [4/]
														--	--	X		3.4.2 and Table IV
Figure 1	MIL-11018-M	X	X	X	X	X	X	--	DCEP	Flat	X	--	X	X	X	--	MIL-DTL-22200	As-welded condition (3.4.1 and Table III) Stress-relieved condition [4/]
														--	--	X		3.4.2 and Table IV
Figure 2	MIL-8018-C3	X	X	X	--	--	--	--	DCEP [6/]	Vertical	X	--	X	--	--	--	MIL-DTL-22200	As-welded condition
Figure 2	MIL-8018-C3SR	X	X	X	--	--	--	--	DCEP [6/]	Vertical	X	--	X	--	--	--	MIL-DTL-22200	Stress-relieved conditions [5/]
Figure 2	MIL-9018-M MIL-11018-M	X	X	X	--	--	--	--	DCEP [6/]	Vertical	X	--	X	--	--	--	MIL-DTL-22200	As-welded condition Stress-relieved condition [4/]

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

TABLE VIII. Groove weld tests required for conformance inspection – Continued.

NOTES:

1/ All NDT inspection (VT, RT, and MT) of weld test assemblies shall be performed to the procedures and requirements of MIL-DTL-22200. Acceptance criteria shall be in accordance with MIL-DTL-22200.

2/ Two tensile tests shall be prepared and tested, and includes ultimate, yield, elongation, and reduction in area.

3/ If chemical analysis is not determined using chemical analysis test pads welded in accordance with MIL-DTL-22200, chemical analysis shall be determined from milling sample taken from the centerline of the broken tensile specimens or drillings, or by optical emission spectroscopy on any sample that is taken and analyzed at the depth of the tensile specimen centerline. In event of variances with composition requirements or conflicting results by testing activities, analysis shall be verified by procedures in accordance with MIL-DTL-22200.

4/ When specified (see 6.2), types MIL-7018-M, MIL-9018-M, and MIL-11018-M shall be conformance inspected in the stress-relieved condition only, as specified in 3.4.1.a or 3.4.1.c, as applicable. The following suffixes shall be added to the designation when tested in the stress-relieved condition.

SR1 – Stress-relieved in accordance with 3.4.1.a.

SR3 – Stress-relieved in accordance with 3.4.1.c.

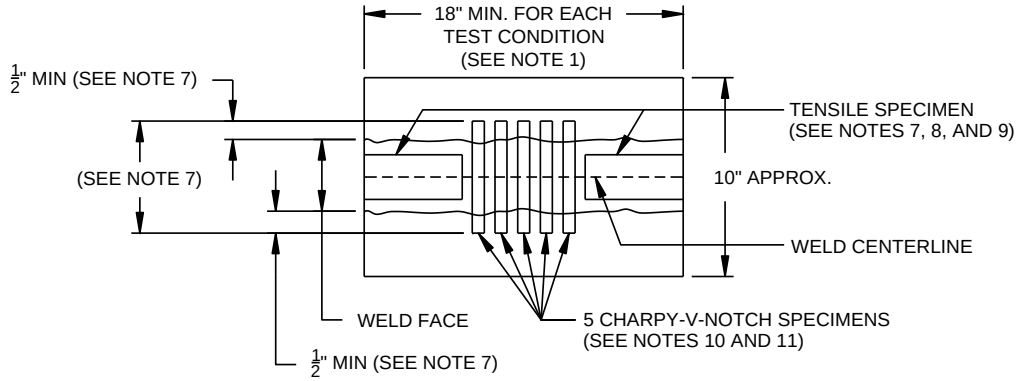
5/ Type MIL-8018-C3SR shall be tested in the stress-relieved condition only using treatment as specified in 3.4.1.a or 3.4.1.b (see 6.2). The following suffixes shall be added to the designation when tested in the stress-relieved condition.

SR1 – Stress-relieved in accordance with 3.4.1.a.

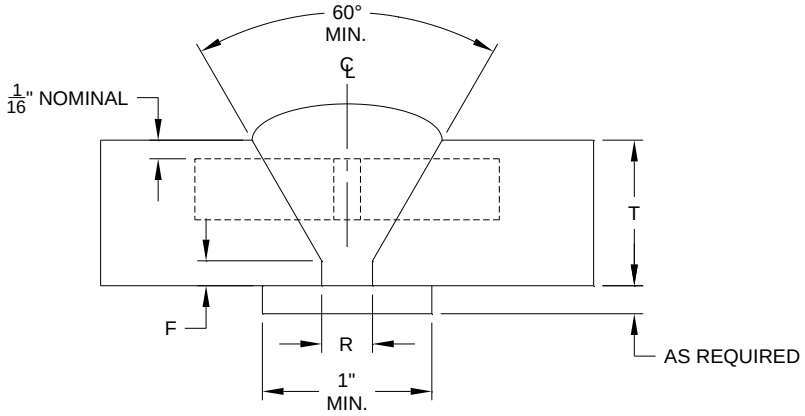
SR2 – Stress-relieved in accordance with 3.4.1.b.

6/ Unless otherwise specified (see 6.2), tests shall be done with DCEP only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.



LAYOUT OF A TEST PLATE SHOWING THE LOCATION OF TEST SPECIMENS



GROOVE PREPARATION OF A TEST PLATE
SHOWING THE ORIENTATION OF THE CHARPY-V TEST SPECIMEN

NOTES:

1. Each test condition refers to as-welded or stress-relieved conditions.
2. Base plate material shall be as specified in 4.4.1. See table IX for required weld joint dimensions for all MIL-types except MIL-11018-M. Unless otherwise specified (see 6.2), refer to table X for MIL-11018-M weld parameters and groove weld preparation requirements.
3. Welds shall be deposited in approximately ⅛-inch-thick layers. The weld bead layers shall be deposited with a maximum weave width (electrode manipulation) of 2½ times the core wire diameter. The final layer shall include a reinforcement of standard proportions (1/32 inch to ⅛ inch). Welding parameters shall be as specified in the manufacturer's weld procedure. Each layer shall be started at the finishing end of the preceding layer. This note does not apply to MIL-11018-M and MIL-7018-M electrodes.
4. For MIL-7018-M electrodes, the groove preparation of the test plate shall be a 45-degree included angle; the backing strap shall be ½-inch nominal thickness by a minimum width of 1 inch; the test plate width shall be a minimum of 15 inches and the length shall be a minimum of 18 inches; the heat input shall be 50 to 60 kilojoules per inch (kJ/in) for the ⅛-inch-diameter and larger electrodes and 30 to 40 kJ/in for the 3/32-inch-diameter electrode; and the split-weave technique (the practice of depositing only the two wide-weave beads per weld layer) shall be limited to the first ½T of the weld joint thickness, and each weld layer beyond the first ½T of weld joint thickness shall contain three or more weld beads. The maximum reinforcement for the final layer shall not exceed 3/16 inch.

FIGURE 1. Weld joint for mechanical and impact tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

NOTES (continued):

5. After the joint has been tacked, the assembly shall be preheated to 225±25 °F prior to welding. After completion of each pass, the interpass temperature of the assembly shall be checked and the assembly shall be allowed to cool or shall be reheated, as required to maintain the assembly temperature in the range of 225±25 °F. Any required cooling of test assemblies shall be done in still air at room temperature, and assemblies shall not be force cooled with compressed air or by any other means. If it is necessary to interrupt this procedure, and the assembly cools, the assembly shall be reheated to 225±25 °F prior to resuming welding. When specified (see 6.2), the minimum preheat and interpass temperature shall be 60 °F when using MIL-7018-M electrode for welding HSLA-80 or HSLA-100 steel. This note does not apply to MIL-11018-M electrodes.
6. After completion of the weld and VT inspection is conducted for the as-welded condition, the weld reinforcement and backing strips shall be removed flush with the base plate on both surfaces and the assembly shall be subjected to the additional required nondestructive (NDT) inspections. Weld pickup (i.e., welding performed prior to formal NDT inspection to correct minor visual surface irregularities, such as low areas in the weld reinforcement, weld bead roll and a start-stop area) and minor repair (e.g., shallow lack of fusion against backing bar) may be accomplished on the weld face (before or after flush grinding or machining) and on the backside after backing bar removal provided that the same electrode being tested is used in conjunction with the same welding conditions, welding position, and other controls used in the original welding of the test assembly. See note 12 for the stress-relieved condition.
7. The weld shall be cut as shown resulting in two tensile coupons (i.e., blanks) for 0.500-inch-diameter specimens and a center coupon of size sufficient for the required Charpy V-notch specimens. No base plate material shall be removed within ½ inch of the edges of the face of the weld by flame cutting. Only sawing or machining shall be used for specimen finishing.
8. One tensile coupon, where required, shall be removed from the stress-relieved welded test assembly as specified in 3.4.1 herein. The centerline of the all-weld-metal tensile specimens (as-welded and stress-relieved) shall be nominally ⅜ inch below the top surface (or nominally ¼ inch below the surface for ½-inch-thick welds) of the test plate.
9. Both the stress-relieved and as-welded coupons, when required, shall be machined into 0.500-inch tensile specimens. For details see MIL-DTL-22200.
10. The center section of the welded joint shall be used for the preparation of Charpy V-notch impact specimens. The Charpy V-notch specimens shall be removed and machined in accordance with MIL-DTL-22200 and to the details shown on this figure with the exception that on the weld joints welded with the MIL-7018-M electrode the Charpy impact specimens shall be taken from a location nominally ⅛-inch below the surface. Prior to machining specimens, the welded test assembly shall be stress-relieved as specified in tables VII and VIII, as applicable.
11. Impact properties for the five Charpy V-notch specimens for each test condition shall be obtained at test temperatures as specified in table III. The computed average value shall be determined as specified in the notes to table III and shall be greater than or equal to that as specified in table III.
12. When testing in the stress-relieved condition is required, VT inspection of the weld shall be conducted prior to stress relief. Subsequent to VT inspection, the weld assembly shall be stress-relieved prior to removing test coupons or removing the backing strap. Subsequent to the stress-relief heat treatment, the weld assembly shall be allowed to cool and the weld reinforcement and backing straps shall be removed flush with the base plate on both surfaces and the assembly subjected to the additional required NDT inspections.

FIGURE 1. Weld joint for mechanical and impact tests – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

TABLE IX. Groove weld preparation. [1/, 2/]

Electrode size (inch)	Minimum thickness (T) (inches)	Maximum root opening (R) [3/] (inch)	Root face (F) (inch)
3/32	3/4	1/4	0 to 1/32
1/8	3/4	1/4	0 to 1/32
5/32	3/4	1/4	0 to 1/32
3/16	3/4	1/4	0 to 1/32
7/32	3/4	1/2	0 to 1/32
1/4	1	1/2	0 to 1/32
5/16	1 1/4	1/2	0 to 1/32

NOTES:

1/ Does not apply to MIL-11018-M electrodes.

2/ Pass and layer sequence shall be recorded. A sketch is acceptable, as is use of written format for bead placement, such as 1 full, 2 split, 3 triple, and 5 quad layers.

3/ Root opening may be increased to allow proper bead placement and avoid slag entrapment.

TABLE X. MIL-11018-M welding requirements. [1/, 2/]

Electrode size (inch)	Minimum thickness (T) (inch)	Maximum root opening (R) (inch)	Root face (F) (inch)	Welding parameters	
				Heat input (kJ/in) [3/, 4/]	Preheat/interpass temperature (°F) [5/, 6/]
3/32	3/4	1/2	0 to 1/32	30 to 35	200 to 225
1/8	3/4	1/2	0 to 1/32	35 to 41	200 to 225
5/32	3/4	1/2	0 to 1/32	45 to 51	200 to 225
3/16	3/4	1/2	0 to 1/32	51 to 57	200 to 225
7/32	1	1/2	0 to 1/32	51 to 57	200 to 225
1/4	1	1/2	0 to 1/32	51 to 57	200 to 225

NOTES:

1/ Pass and layer sequence shall be recorded. A sketch is acceptable, as is use of written format for bead placement, such as 1 full, 2 split, 3 triple, and 5 quad layers. The final weld layers shall include a reinforcement of standard proportions (1/32 to 1/8 inch).

2/ Welded test assemblies shall be insulated or separated from the workbench. Unless otherwise specified (see 6.2), to minimize the angular distortion of the completed test weld, test assemblies shall be fully restrained during heating, welding, and cooling by appropriate fixturing. Restraining fixtures shall not interfere with the placement of strip heaters or heating pads on the backside of the assembly. Contact points between the fixture and the test plates shall be limited to minimize the heat sink effect of the fixturing. Base plates either side of the weld joint shall not be back set (reverse distorted) before welding to compensate for weld shrinkage, as an alternative to the required fixturing.

3/ Alternate NAVSEA-approved heat inputs and test plate thickness may be specified (see 6.2).

4/ See 4.4.2 for heat input calculation procedures.

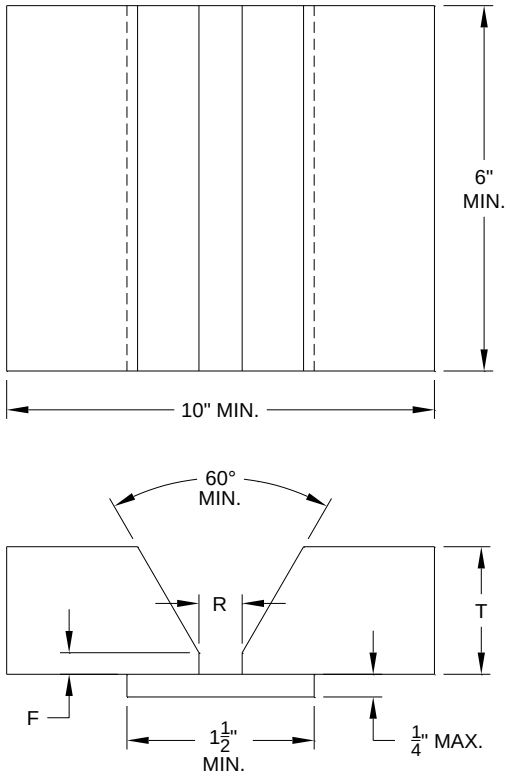
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

TABLE X. MIL-11018-M welding requirements – Continued. [1/, 2/]

NOTES:

5/ Preheat shall be applied evenly along the length of the plate using electric resistance contact strip heaters or electric resistance ceramic pad heaters. If temperature indicating crayons are used, the preheat and interpass temperature shall be measured at several points along the entire length of weld 1 to 1½ inches from the edge of the bevel on both plates being joined. If a contact pyrometer is used, the preheat and interpass temperature shall be measured in at least 3 places 1 to 1½ inches from the bevel on each side of the plate (at each end and the middle). The heat shall be applied in a way to ensure that it is through thickness, with the heater strips or heater pads on the bottom of the plate (opposite from the side being welded).

6/ After completion of each pass, the interpass temperature of the assembly shall be checked and the assembly shall be allowed to cool or shall be reheated, as required to maintain the assembly temperature in the preheat range as specified in this table. Any required cooling of test assemblies shall be done in still air at room temperature, and assemblies shall not be force cooled with compressed air or by any other means. If it is necessary to interrupt this procedure, and the assembly cools, the assembly shall be reheated to the preheat temperature, as specified in this table, prior to resuming welding.



NOTES:

1. Welding current shall be as specified in the manufacturer's weld procedure.
2. Base plate material shall be as specified in 4.4.1. See table XI for required weld joint dimensions.
3. If flame cut, edge preparation shall be ground to remove slag and surface irregularities.
4. Welds shall be deposited in approximately ⅛-inch-thick layers. Weld bead widths shall be as shown in table XI. Each pass shall contain a start in the area to be evaluated.

FIGURE 2. Groove weld usability test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

NOTES (continued):

5. The as-welded reinforcement shall be a maximum of 3/16-inch.
6. Preheat and interpass temperature shall be 225±25 °F.
7. After completion of the weld and conduct of VT inspection, the weld reinforcement and backing strips shall be removed flush with the base plate on both surfaces and the assembly shall be subjected to the additional required NDT inspections. Weld pickup and minor repair may be accomplished on the weld face (before or after flush grinding) and on the backside after backing bar removal provided that the same electrode being tested is used in conjunction with the same welding conditions, welding position, and other controls used in the original welding of the test assembly.
8. Not applicable to MIL-7018-M.

FIGURE 2. Groove weld usability test – Continued.

TABLE XI. Groove weld usability test preparation.

Measurements	Electrode size (inch)		
	3/32	1/8	5/32
Thickness (T)(inch)(minimum)	1/4	3/8	1/2
Root open (R)(inch)	3/16 to 5/16	1/4 to 3/8	5/16 to 7/16
Root face (F)(inch)	0 to 1/32	0 to 1/32	0 to 1/32
Weld bead width (maximum)(inch)	3/8	1/2	5/8

4.4 Procedures and methods.

4.4.1 Weld joint base plate material. Ordinary-strength hull structural steel or higher-strength hull structural steel base plate material in accordance with MIL-S-22698 and ASTM A36/A36M plate may be used for all electrode tests, except for MIL-11018-M electrodes. For MIL-11018-M type electrodes, the base plate material shall be HY-80 steel in accordance with T9074-BD-GIB-010/0300. The plate shall meet the chemical composition requirements of T9074-BD-GIB-010/0300 and can be in the as-rolled condition without being quenched and tempered or tested for mechanical properties. For MIL-7018-M, when specified (see 6.2), the base plate material shall be HSLA-80 or HSLA-100 steel in accordance with T9074-BD-GIB-010/0300, and the preheat and interpass temperature shall be a minimum of 60 °F.

4.4.2 Calculation of welding heat input. Heat input (J/in) shall be calculated using the following formula:

$$\text{Heat Input (J/in)} = \frac{\text{Volts x Amperes x 60}}{\text{Travel Speed (inches per minute)}}$$

The root pass(es) shall be excluded from all heat input calculations. Specified heat inputs reflect minimum acceptable ranges and qualification or conformance testing at values beyond the specified range shall be accepted.

4.4.2.1 Operational heat input calculation. Operational heat input shall be the actual heat input recorded or calculated for individual passes during welding of the test assemblies. For shielded metal arc welding, calculate the operational heat input of each pass in the test assembly using the following procedure:

a. To determine the travel speed for each complete weld pass, divide the total length of the complete weld pass by the total arc time for the pass. Total length of the complete weld pass and total arc time for the pass shall be calculated as in b and c below, respectively.

b. On MIL-11018-M to determine the total length of the complete weld pass, measure each weld bead individually and add the bead lengths. Measure each bead from the apparent trailing edge of the beginning of the bead to the apparent trailing edge of the stop crater. On other electrode types the total completed weld pass may be

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

measured to determine length, or alternately, the method described for MIL-11018-M may be used. Do not use the length of the test assembly as the pass length.

c. To determine the total arc time of the complete weld pass, measure the arc time for each weld bead in the pass and add the individual bead arc times together. An electronic arc timer or equivalent automated system that measures the total arc time for the pass may also be used.

d. Measure the arc voltage and welding amperage for each weld bead in the pass. Average the voltage and amperage to derive the average voltage and amperage values for the weld pass. If no adjustments are made to the output of the welding equipment, it is not necessary to measure the amps and volts on each bead.

e. Insert the average travel speed, arc voltage, and amperage for each complete pass into the heat input equation of 4.4.2.2 to calculate the heat input for the weld pass.

4.4.2.2 Average heat input. The average heat input shall be the average of the operational heat input of all passes in the test assembly and shall fall within the range (and tolerances) specified in the notes to figure 1 and table X.

4.5 Temperature and humidity environmental exposure procedures. The electrode sample shall be exposed to elevated temperature and humidity conditions in accordance with exposure procedures in accordance with AWS A5.5/A5.5M. At the end of the exposure time, the electrodes shall be removed from the chamber and tested for covering moisture content in accordance with MIL-DTL-22200 (see table II, note 2).

4.5.1 Frequency of testing. The test of 4.5 shall be conducted for each lot, except that after testing ten consecutive lots of a specific product and size electrode with satisfactory results, the frequency of testing for that specific product and size electrode may be reduced to one lot in every ten lots. However, one unsatisfactory test result shall require reversion to the above cycle.

4.6 Inspection of packaging and marking. Inspection of packaging and marking shall be in accordance with MIL-DTL-22200.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The range of electrode types covered are several low-hydrogen types of arc-welding electrodes of various alloys and strength levels, capable of depositing weld metal that, in the as-welded or stress-relieved conditions, meets mechanical property requirements given herein. It will be noted that, although these electrodes may be used for some applications in the as-welded condition, they are capable of stress relief as required. Properties may be affected by long time stress relieving and slow cooling from stress-relieving temperature. Properties should be determined to be satisfactory after a simulated production postweld heat treatment by the user when required by the fabrication document. The designation "low-hydrogen" applies to electrodes having coverings that are substantially free of any compound that may liberate hydrogen in the arc. Low-alloy steel electrodes that produce low-hydrogen in the arc have been found suitable for welding many steels that formerly gave serious difficulty through excessive porosity and underbead cracking, unless welded with high-alloy electrodes. Intended use of each type is as specified in 6.1.1 through 6.1.6. Current stock of electrode types specified in this specification may be used until depleted; future requisitions for electrode types specified in this specification should refer to the current version of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

6.1.1 MIL-7018-M. This type is intended to cover medium steel electrodes used in welding medium or high-tensile steels, and to attach such steels to high yield strength steels such as HY-80/100 and HSLA-80/100. This electrode is intended for use with DCEP electrical characteristics only.

6.1.2 MIL-8018-C3. This type is intended to cover low-alloy electrodes capable of depositing weld metal of higher strength than type MIL-7018-M.

6.1.3 MIL-8018-C3SR. This type is intended to cover a special MIL-type electrode formulated to meet the long time (50 hours) stress-relieved mechanical properties requirements only.

6.1.4 MIL-9018-M. This type is intended to cover low-alloy electrodes suitable for welding steels having properties similar to those of the weld, and for welding attachments to HY-80 steel, as may be permitted. This electrode is intended to be used with the DCEP electrical characteristics only.

6.1.5 MIL-11018-M. This type is intended to cover low-alloy electrodes for welding groove butt joints in HY-80, HSLA-80, HSLA-100, and HY-100 steels where approved. This electrode is intended to be used with DCEP electrical characteristics only.

6.1.6 All types with suffix RC. This type is intended to cover electrodes for welding pressure and strength welds that will be subject to irradiation.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Type, class, and size required (see 1.2).

c. Whether stress-relieving treatments are required (see 3.4.1 and table VIII).

d. Whether a maximum chromium content of 0.20 percent is required to control hexavalent chromium in welding fumes (see table IV, note 3).

e. Whether the copper content is restricted as designated by the suffix RC to the MIL-type (see table IV, note 4).

f. For type MIL-8018-C3SR, the length required (see 3.5).

g. For 3/32-inch size electrode, whether 9-, 12-, or 14-inch length is required (see 3.5 and table V).

h. For MIL-11018-M, when parameters other than those of table X are to be used (see figure 1, note 2 and table X, note 3).

i. Whether HSLA-80 or HSLA-100 steel and a 60 °F preheat and interpass temperature is used for evaluating MIL-7018-M electrodes (see figure 1, note 5 and 4.4.1).

j. If alternate restraint methods other than fixturing are acceptable to minimize angular distortion of test welds (see table X, note 2).

k. If welding is to be performed using other than DCEP (see table VIII, note 6).

l. For MIL-8018-C3SR, the stress-relieved condition required for conformance inspection (see table VIII, note 5).

m. Whether types MIL-7018-M, MIL-9018-M, or MIL-11018-M are to be conformance inspected in the stress-relieved condition (see table VIII, note 4).

n. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 22200 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.

products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Subject term (key word) listing.

Heat input
Hydrogen, diffusible
MIL-7018-M
MIL-8018-C3
MIL-8018-C3SR
MIL-9018-M
MIL-11018-M
Moisture, coating
Wire, core

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – GS

Preparing activity:
Navy – SH
(Project 3439-2023-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:20Z
Check the source to verify that this is the current version before use.